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Introduction

저는 서울대학교 3학년 1학기 재학 중인 2021-13365 안준현입니다.
저는 통계학, 그래프 이론, 그리고 매듭 이론에 관심을 갖고 공부하고 있습니다.

Education

- 2021
해석개론1(3341.201.001)
해석개론2(3341.202.002)
- 2022-2023
군휴학
- 2024
수리통계1 (326.311.001)
위상수학개론1 (881.401.001)
해석개론 및 연습1 (M1407.000600.001)
수리통계2 (326.312.001)
다변량자료분석 및 실습 (326.316.001)
수치선형대수 (881.319.001)
위상수학개론2 (881.402.001)
- 2025
현대대수학1 (881.423.001)
회귀분석 및 실습 (326.313.001)
편미분방정식 (881.423.001)
측도이론과 확률 (M1407.002500.001)
응용수학특강 (3341.751.002)
베이지스통계 및 실습(326.411.001)
생존자료분석 및 실습(326.412.001)
확률론2 (326.516.001)
대수적위상수학1 (3341.607.001)
대수학특강 (3341.715.002)

Experience

Knot theory, RNE with Jae Ik Park, Jin Wook Kim, and Adv. Gyo Taek Jin May 2019 – Dec 2019

The Smallest Lattice Cube Containing Nontrivial Band Knots

- Algorithm of generating all possible band knot on $n \times n \times n$ unit cube(Python code)
- Proof that all possible band knot under $3 \times 3 \times 3$ unit cube are topologically equivalent to unknot; showing that there is no possible crossing by projection on a sphere
- Existence of möbius strip on $2 \times 2 \times 2$ unit cube and trefoil knot on $4 \times 4 \times 4$ unit cube

Graph theory

May 2020 – Dec 2020,

Feb 2023 - Sep 2023

On the Computational Complexity and Approximation of the Rural Postman Problem in Euclidean Graphs

- Optimal solution occurs when minimal matching of odd vertices makes disjoint graphs connected.
- Proof that adding a double edge between each graphs makes upper bound of optimal solution with triple the additional cost
- Proof that the problem is NP-hard; showing that Euclidean TSP problem is a special case of this problem.
- Using minimal matching, adding double edge of disjoint components, and minimum spanning tree we can find upper bound of optimal solution in polynomial time.